

A Prospective Evaluation of Borehole Distributed Acoustic Sensing Technologies for Detecting and Locating Out-of-Network Microseismicity

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ABSTRACT

Downhole distributed acoustic sensing (DAS) has proven effective in monitoring microseismicity produced during hydraulic stimulations. It is also a promising tool for monitoring microseismicity occurring beyond the stimulation region. Downhole DAS arrays exhibit reduced sensitivity to surface noise compared to surface deployments, and they can remain in place undisturbed after operations cease, unlike downhole geophones. Identifying and locating microseismicity occurring at remote distances is crucial for reservoir monitoring, allowing us to assess stress conditions and the presence of pre-existing faults. Here, we evaluate the potential of downhole DAS in detecting and locating out-of-network microseismicity through two case studies: a field-scale investigation at the Frontier Observatory for Research in Geothermal Energy (FORGE) site in Utah and a mesoscale experiment in the Bedretto Underground Laboratory for Geosciences and Geoenergies (BULGG) in Switzerland. We demonstrate that DAS data have sufficient quality to detect and locate low-magnitude earthquakes ($M < 0$) occurring several hundred meters away. However, azimuthal ambiguity is unavoidable when only a single vertical borehole is available. We show that DAS signal characteristics and location capabilities perform equally at different scales, ranging from full-reservoir scale to mesoscale. Our evaluation supports the employment of longer-term experiments to explore downhole DAS suitability for passive microseismic monitoring.

16.1 INTRODUCTION

Several industry operations, from petroleum to geothermal applications, rely on locations of microseismic events to gain insight into underground processes. The goal of microseismic data processing is to transform continuous wavefield observations into precise and accurate estimates of event locations and magnitudes and other source characteristics. Knowledge of such parameters is key for surveillance and diagnostic characterization of induced seismicity both in the context of the development of the reservoir connectivity (e.g., reconstruction of spatiotemporal fracture development) and for safety protocols and risk governance (Chen et al., 2019; Grigoli et al., 2022; Van Der Baan et al., 2013). Typical microseismic monitoring systems use both a permanent network of sensors deployed at the surface and/or in shallow boreholes and deep downhole geophone arrays as close as possible to the stimulation region (Eaton, 2018).

Recently, distributed acoustic sensing (DAS) technology has become available, turning fiber-optic cables (e.g., conventional telecommunication or built-for-purpose cables) into an ultra-dense network of strain sensing points that can sample the seismic wavefield continuously for tens of kilometers. Existing infrastructures such as telecommunication fiber-optic cables, can be used for distributed sensing. Thus, DAS technology has been increasingly used as a promising and cost-effective subsurface monitoring tool for a wide range of applications across many research fields and environments: from urban seismology (Dou et al., 2017; Song et al., 2022; Spica et al., 2020; Yuan et al., 2020) to glacial studies (Booth et al., 2020; Brisbourne et al., 2021; Fichtner et al., 2023; Hudson et al., 2021), volcanology (Currenti et al., 2021; Jousset et al., 2022; Klaasen et al., 2021; Nishimura et al., 2021), and regional and local earthquake monitoring as well as structural imaging (Biondi et al., 2021a; Biondi et al., 2021b; Jousset et al., 2018; Karrenbach et al., 2019; Li et al., 2021; Lindsey et al., 2017). Borehole fiber-optic seismology where cables are installed along the casing or behind the tubing of wells has also gained attention with successful applications for reservoir and fracture imaging (Lellouch et al., 2019b; Martuganova et al., 2022; Staněk et al., 2022; Titov et al., 2022) and microseismic monitoring for different industrial operations, including oil and gas exploration (Jiang et al., 2016; Shultz & Simmons, 2019; Zhan & Nahm, 2020), CO₂ sequestration monitoring (Correa et al., 2019; Daley et al., 2013; Götz et al., 2018; Humphries et al., 2015; Wilson et al., 2021), hydraulic stimulation in unconventional gas exploitation (Bakku, 2015; Karrenbach et al., 2017; Liu et al., 2022; Luo et al., 2021), and enhanced geothermal system (EGS) sites (Lellouch et al., 2020; Lellouch et al., 2021).

In particular, for geothermal systems, DAS technology offers an advantage over traditional seismic sensors for microseismic monitoring because it does not have the same temperature limitations as standard electronic equipment. The sensing fiber can therefore be installed at high-temperature reservoir conditions. The maximum temperature at which fibers can maintain satisfactory performance depends on the materials used for fiber coating; typically, these values range between 300 and 400 °C (Mellors et al., 2018). This increased durability and resistance to corrosion of fiber-optic cables (Reinsch et al., 2015) is also beneficial to other applications where pressure, salinity, or high acidity are elevated. We note, however, that fiber-optic cable deployments in borehole settings are still challenging and are not devoid of practical issues. These challenges include safety concerns during installation, cable breakdowns, management of lengthy cables, and environmental conditions such as temperature and humidity, which may degrade the integrity of interrogator units. When deployments are successful, seismic DAS cables installed at target depth reservoirs can provide ideal conditions to investigate the behavior of the subsurface during stimulation and

utilization. Consequently, the distance between the detecting sensor and the microseismic events can be minimized, thus maximizing the detection capability. In addition, DAS has the potential for recording many more microseismic events than conventional surface arrays or sparse downhole geophone arrays due to its dense spatial coverage over long distances (typical DAS acquisition samples the wavefield at meter or submeter spacing and sampling frequencies of 1 kHz or higher; Lellouch et al., 2020), which can also be used to image subsurface structures at high-resolution (Lellouch et al., 2019a; Staněk et al., 2022). On the other hand, as DAS records the strain or strain rate only along the fiber axial direction, a primary limitation is its cylindrical symmetry, which creates azimuthal ambiguity, especially in purely vertical boreholes and single-borehole monitoring configurations (Karrenbach et al., 2017; Lellouch & Biondi, 2021).

Many studies have tackled the problem of earthquake detection and location with DAS. For surface earthquake monitoring studies, DAS technology demonstrated promising outcomes when traditional detection and location algorithms were employed (e.g., Jousset et al., 2018). However, early applications of conventional detection and location algorithms to DAS data in borehole settings were relatively unsuccessful and underperforming, yielding fewer detections and greater location errors compared to conventional geophones (Hull et al., 2017; Webster et al., 2016). Thus, to fully exploit the high spatial and temporal sampling of DAS data, as well as to deal with the large data volumes (up to several TB per day) of inherently noisier recordings, new processing techniques have been proposed across several different fields of application. For example, the PoroTomo DAS array at Brady Hot Springs geothermal field in Nevada has provided a rich DAS dataset and testbed for template matching techniques (Li & Zhan, 2018), evaluation of beamforming potential in DAS data (van den Ende & Ampuero, 2021), as well as subsurface mapping and seismic event monitoring (Piana Agostinetti et al., 2022). Template matching and beamforming have also been used for earthquake aftershock detection (Li et al., 2021), small earthquake detection at regional distances (Nayak & Ajo-Franklin, 2021), and identifying and locating volcanic events (Jousset et al., 2022). In recent years, machine learning models are being adapted to and trained with continuous data recorded on DAS arrays for earthquake detection (Huot et al., 2018a; Huot & Biondi, 2020), earthquake classification (Currenti et al., 2023), seismic arrival-time picking (Zhu et al., 2023), as well as traffic and infrastructure monitoring (Huot et al., 2017; Huot et al., 2018b; Huot & Biondi, 2018; Martin et al., 2018).

Specifically in the field of microseismic monitoring with downhole DAS, significant advances have been made toward enhancing and/or tailoring detection and location algorithms to conform with DAS data (Lellouch & Biondi, 2021). In their work, Lellouch et al. (2020), using a single vertical fiber, proposed a beamforming approach targeted at event detection and event location without azimuthal information. Machine learning models for microseismic event detection have also shown promising progress, as evidenced by successful applications (e.g., Binder & Chakraborty, 2019; Stork et al., 2020). More recently, deep learning approaches have also emerged, demonstrating significantly enhanced accuracy and outcomes (Huot et al., 2022). Furthermore, Porras et al. (2024) proposed a fast semblance-based seismic event detection method in which waveform coherence is used as the detector threshold, overcoming the high self-noise of DAS records by exploring all the channels at once. The near-real-time performance of the proposed detector has the potential to unlock valuable information during underground operations (e.g., hydraulic fracturing, fluid injection, circulation, and extraction). Tuinstra et al. (2024) tackle the location ambiguity of a single vertical DAS cable by using distance geometry solvers to retrieve the shape and relative location of clustered microseismicity. The method is adapted from the single-station location algorithm for clustered earthquakes (earthquake locAtion via Distance gEometry Solvers [HADES]) described in Grigoli et al. (2021), and it was used to locate off-shore clustered seismicity (Cesca et al., 2021).

Downhole DAS deployments have proven effective in monitoring microseismic events generated during hydraulic stimulation operations and in the proximity of the well being stimulated. However, it has been shown that they can also detect distant earthquakes and often outperform a surface-based regional seismic network by an order of magnitude (Lellouch et al., 2021). This is due to their deeper deployment, which allows bypassing the highly dissipative heterogeneous shallow layers of the Earth and lower sensitivity to surface noise sources such as anthropogenic activities and wind. The inherent lower signal-to-noise ratio (SNR) of individual DAS channel recordings, especially at far-offsets due to the narrower directivity patterns (Correa et al., 2017), can be overcome by beamforming and then, once time delays have been properly corrected for, stacking all the available channels. The ability to detect and locate distant microseismicity is paramount to reservoir monitoring both during and after the end of operations, as it allows us to assess the geomechanical conditions, seismogenic processes, and potential for reactivation of pre-existing, favorably oriented faults that could affect the stimulation region.

In this chapter, we evaluate the potential of DAS technology in out-of-network microseismicity contexts. Here we define seismic events as “out-of-network” when they exhibit an azimuthal gap exceeding 180°, where the gap is defined as the largest angle between two recording stations as seen from the epicenter. We present two case studies in which DAS cables were installed in boreholes and recorded distant seismicity at both the full-reservoir scale and mesoscale, respectively. We analyze the signal characteristics and DAS-based locations of a seismicity cluster recorded ~3 km east of the Frontier Observatory for Research in Geothermal Energy (FORGE) site, Utah, during the 2019 pilot stimulation. Next, we apply the same methods to a seismicity cluster recorded in 2021 ~ 900 m southeast of the Bedretto Underground Laboratory for Geosciences and Geoennergies in Switzerland (BULGG). Leveraging the different scales, we compare performance in terms of data quality, SNR, detectability, and applicability of the HADES-R location methodology (Tuinstra et al., 2024) for the analysis of out-of-network microseismicity in borehole settings.

16.2 METHODS

16.2.1 Preprocessing of DAS Signal

DAS recordings at each sensing point have low sensitivity and generally lower SNRs than downhole geophones. This has been observed in several studies (e.g., Correa et al., 2017; Lellouch et al., 2020), where a direct comparison with colocated geophones highlighted the inherent noisier data quality of DAS data. It is thus usually recommended that a preprocessing step, aimed at denoising the DAS raw data, is performed before further processing. It has been shown that the removal of the mean and linear trends of each trace of the array (linear and mean filtering step) is essential, as it removes most of the interrogator imprint (laser drift, reference loop vibration, etc.). This can be followed by a bandpass filter in the frequency range of interest, where the energy

signal is generally concentrated. Further coherent noise can be removed by applying a frequency-wavenumber (FK) filter. FK filters are often used in active seismic exploration (Duncan & Beresford, 1994), and both traditional and adaptive methods have been applied to DAS data (e.g., Isken et al., 2022). In the following examples, a traditional FK filter is applied along the $k = 0$ wavenumber to remove noise generated by the interrogator, and, for the FORGE datasets, only the upgoing part of the wavefield was preserved to also remove downgoing free-surface reflections.

16.2.2 HADES-R Location Algorithm

The one-component nature of a vertically deployed DAS cable can lead to difficulties when locating events. When only a single cable is available, there is no information on the azimuth of the seismic event unless prior information is provided (Karrenbach et al., 2019; Lellouch et al., 2021). To overcome some of these challenges, Tuinstra et al. (2024) put forward a new methodology to locate clustered microseismic events with a fiber-optic cable deployed downhole in nonideal network configurations. The method builds on the work of Grigoli et al. (2021), which proposed an efficient method based on distance geometry problem (DGP) solvers for locating clustered seismicity with sparse surface networks of one or two stations (HADES). The method assumes that the events to be located belong to the same cluster and that this cluster is smaller than the distance between the receivers and the cluster. This is a condition that is easily satisfied in an out-of-network setting.

The new version of HADES (HADES-R) described in Tuinstra et al. (2024) can reconstruct the shape and relative location of a seismicity cluster using P- and S-wave arrival times and one a priori located event. To find event locations in a relative frame, HADES-R uses the inter-event distance between two events computed from the double difference in arrival times of S and P phases (t_s and t_p) and measured at a seismic station (equation (16.1)):

$$\|r_{e1e2}^s\| = k_v \left| \left(t_s^{e1} - t_p^{e1} \right) - \left(t_s^{e2} - t_p^{e2} \right) \right| \quad (16.1)$$

where r^s is the inter-event distance between event $e1$ and event $e2$, and $k_v = \frac{v_p v_s}{(v_p - v_s)}$.

In Grigoli et al. (2021) a minimum of four master events with known locations was required to find the absolute location and rotation of the events in the cluster. Any rotational error of the cluster was minimized in the horizontal (latitude–longitude) plane, whereas rotational errors in depth remained unresolved. The main advantage of this method lies in the ability to provide the correct shape of clustered seismicity, particularly in ill-posed event-receiver configurations. In Tuinstra et al. (2024), the master event dependency is reduced from four to one, and multiple random combinations of DAS channels are considered instead of the original two station condition. Once a master event is selected (e.g., a perf shot, the largest event in the sequence), it represents the start of a coordinate system, and three other reference events are then located relative to it to construct a relative, arbitrarily oriented coordinate system. It is not necessary to know the absolute location of the reference events. However, an estimate of the width of the reference frame is required.

As the cluster's initial location is derived from arbitrary-located reference events, a rotation step is necessary to determine the cluster's absolute location. The rotation of the cluster is found by using quaternions, which can describe rotations around arbitrary axes in higher-dimensional space, specifically using a four-dimensional representation (Shoemaker, 1985). The cluster rotation procedure consists of searching for the correct distribution of observed arrival times with respect to distance to the receivers (e.g., the DAS cable). To this end, the master event serves as the rotational origin of the seismic cluster, and three rotation optimization axes are defined: (a) a vertical axis from the origin pointing upward to the surface, (b) an orthogonal axis from the origin pointing to one selected station on the horizontally projected plane, and (c) the cross product of the first two axes. The choice of station that governs axis (b) can be interchanged. The optimal rotation angle for each of the three axes is then the one that maximizes the rectilinearity of the absolute distance between the receiver and the events with respect to the differential traveltime $t_s - t_p$ at each receiver. Ideally, the distribution of $t_s - t_p$ is linearly increasing with increasing distance to the receivers. We caution, however, that in some angular situations, the solution to the rotation problem may be nonunique: that is, multiple rotations of the cluster will result in equivalent distance-traveltime distributions. This can be the case of single vertical boreholes, where lateral rotations could result in equivalent traveltimes. Despite the challenges posed by these unfavorable network configurations, the cluster's overall shape remains intact. Although the method does not fully resolve the azimuthal ambiguity in such cases, the abundance of channel pairs along the length of the DAS cable enables the application of an iterative bootstrap approach and thus allows us to retrieve location uncertainties. For each channel pair, we can apply HADES-R followed by a rotation optimization. The ensemble of solutions from all the channel pairs can aid in constraining uncertainties and identifying possible biases or ambiguities in azimuthal orientation. Ensemble solutions are typically shown as two-dimensional histograms of the output of multiple iterations of HADES-R and the associated rotation step conducted over multiple channel pairs.

As the input data for the method consists of P- and S-wave arrival times, Tuinstra et al. (2024) also propose a semiautomated procedure that allows us to pick efficiently the large volume of data usually recorded by the DAS systems. (Figure 16.1(a)–(c)). P- and S-wave arrivals are initially manually identified on traces with strong SNRs. Subsequently, a user-defined search range is selected encompassing the initial arrivals. Typically, the search area for the S-phase arrival is broader than that used for P-phase arrivals. Finally, the P and S master traces, generated by stacking the manually selected arrivals, are cross-correlated within the predefined search range. This process automatically identifies arrivals in all traces. This is just one of the methods to retrieve P and S picks. Several methods exist in the literature, from semiautomated to semisupervised machine learning methods, to address the phase-picking task for the ultra-dense spatial sampling of DAS data (e.g., Zhu et al., 2023).

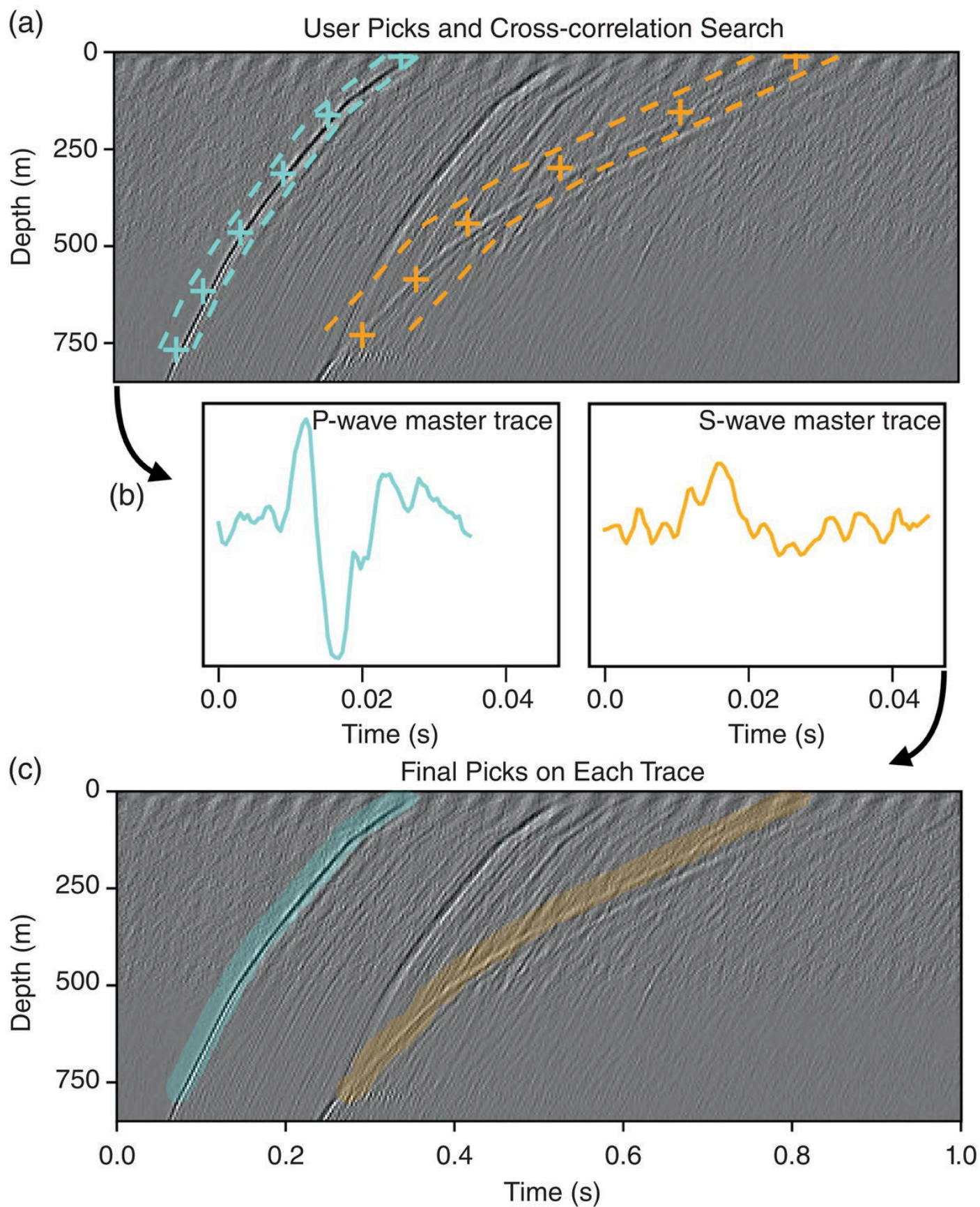


Figure 16.1 Semiautomated workflow for picking P- and S-wave arrivals for DAS data. (a) Manual picks for P- and S-wave arrivals (crosses) and user-defined search range (dotted lines). (b) Master trace derived from stacking manually picked traces over the predefined search range. (c) Final automatic picks for all traces after cross-correlation.

Adapted from Tuinstra et al. (2024).

16.3 CASE STUDIES

16.3.1 UTAH FORGE EGS Test Site

The U.S. Department of Energy FORGE is a dedicated international underground field laboratory in Utah. Its purpose is to develop, test, and accelerate breakthroughs in EGS in a controlled environment. FORGE is located in an unpopulated area on the western flank of the Mineral Mountains in the vicinity of Milford in Beaver County, Utah, a small community of 1,400 people (Figure 16.2; Moore et al., 2020; Wannamaker et al., 2020; Pankow et al., 2020), and close to the active Blundell geothermal plant in the Roosevelt Hot Springs (Gwynn et al., 2016) within Utah's Renewable Energy corridor. FORGE sits on alluvial fan deposits of 300–1,000 m in thickness, with the target EGS reservoir to be created in the underlying basement granitoid rocks where temperatures have been estimated to exceed 190 °C at less than 3 km depth. The site is in continuous development, with new wells being drilled and new instrumentation being added.

The first pilot stimulation took place in 2019 between 21 April and 3 May. At the time, in addition to stations of the Utah Regional Seismic Network (URSN), surface monitoring consisted of five surface broadband stations (red triangles in Figure 16.2); three accelerometers, one placed in a shallow well (68-32); and several temporary high-density nodal arrays in various configurations covering the footprint of the FORGE site. Additional details about the instrumentation in the surrounding vicinity can be found in previous studies (e.g., Pankow et al., 2020; Potter, 2017) as well as on the IRIS website (<https://ds.iris.edu/mda/UU/>). Among the wells drilled at FORGE, Well 58-32 was used for the hydraulic stimulation, and Well 78-32, located about 400 m east-southeast of 58-32, was used for deep seismic monitoring (Figure 16.2). Well 78-32 was equipped with an industry-grade 12-level geophone string deployed by Schlumberger and an optical fiber in a metal tube cemented behind the casing. The geophones were positioned at depths of 645–980 m, with 30.5 m spacing between each sensor, and they were recording only during the stimulation. The installed fiber cable included two multimode (MM) fibers, two single-mode (SM) fibers, and a specialized Silixa Constellation fiber (Diaz-Meza et al., 2023). Fibers were looped in U bends and protected in bottom hole assembly (BHA) housing. The cables were interrogated using the Carina system (Naldrett et al., 2020). Correa et al. (2017) demonstrated that this engineered fiber yields a substantial 20 dB improvement in the optical SNR with respect to recordings using standard fibers. DAS data were acquired with a 1 m channel spacing using a 10 m gauge length and a sampling rate of 2 kHz for a 1,000-m-long cable. Additionally, before being written to disk, the laser sampling rate underwent a 16-fold internal stacking process (Lellouch et al., 2021). This process enables acquiring native data at very high sampling rates to achieve high SNR while avoiding extremely large data volumes by reducing the output frequency.

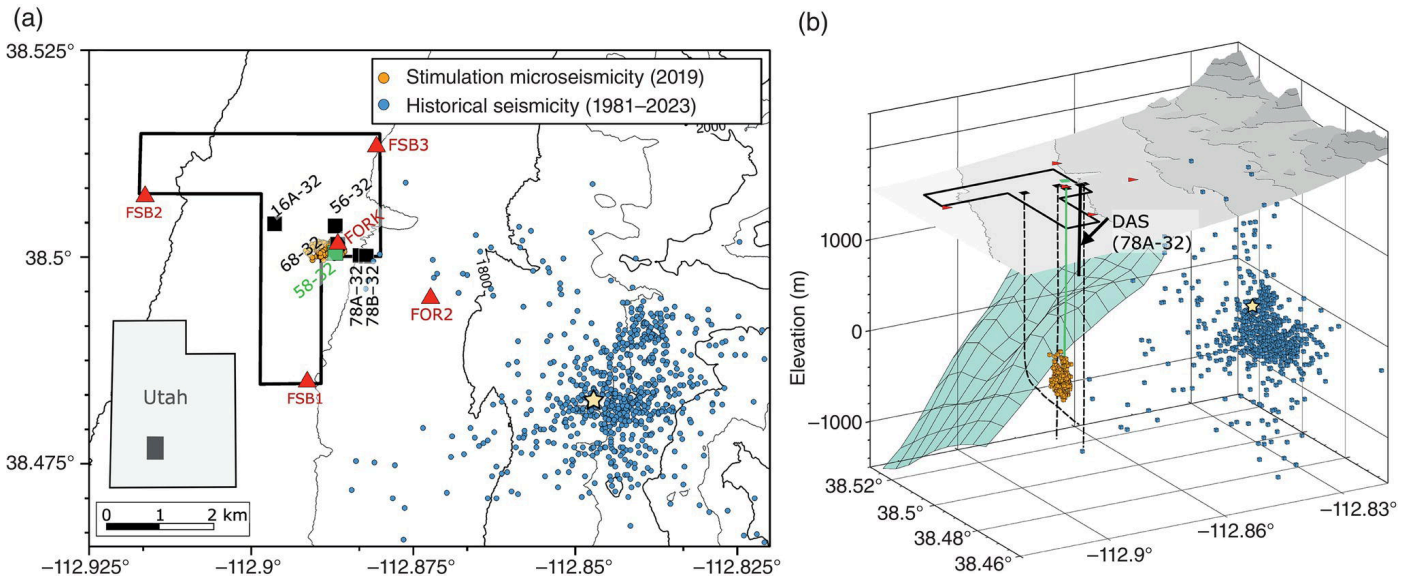


Figure 16.2 (a) Map of the FORGE geothermal test site in Utah, USA. Historical seismicity for the period between January 1981 and March 2023, cataloged by the University of Utah Seismograph Stations, is shown with solid blue circles (this represents what hereafter is referred to as a “southeast cluster”). The microseismicity recorded during the 2019 FORGE pilot stimulation is shown with solid orange circles. The green square illustrates the stimulation well (58-32) used during the pilot stimulation in 2019. All wells drilled to date (78A-32, 78B-32, 68-32, 56-32, and 16A-32) are indicated by black squares, and the seismic stations of the seismic network currently active are shown with red triangles. (b) 3D rendition of the site highlighting the dipping granitoid layer (dark green surface), the pilot injection Well 58-32 (light green line), and the monitoring Well 78-32 instrumented with DAS (solid black line). The yellow star indicates the event shown in Figure 16.3, which was recorded by both DAS and the surface stations.

During the 2019 stimulation, in addition to the near-well microseismicity originated by the stimulated fractures, the DAS fiber installed in the monitoring Well 78-32 recorded a distant cluster of 82 earthquakes located 3–5 and 8–10 km southeast of the stimulation Well 58-32 (Lellouch et al., 2021). Of these, only one event (Figure 16.3) was detected by the routine processing of the regional surface network and reported in the University of Utah Seismograph Stations (UUSS) catalog. This activity is likely not directly associated with the stimulation but rather belongs to a cluster of seismicity well aligned with the historical seismicity in the area, and which is possibly connected to the Mineral Mountains swarms and natural hydrothermal fluid-flow processes (Lellouch et al., 2021; Zandt et al., 1982). In the next sections, we explore the signal characteristics of the proximal subcluster (3–5 km) as recorded by the DAS system and test the DAS location capabilities in challenging out-of-network conditions.

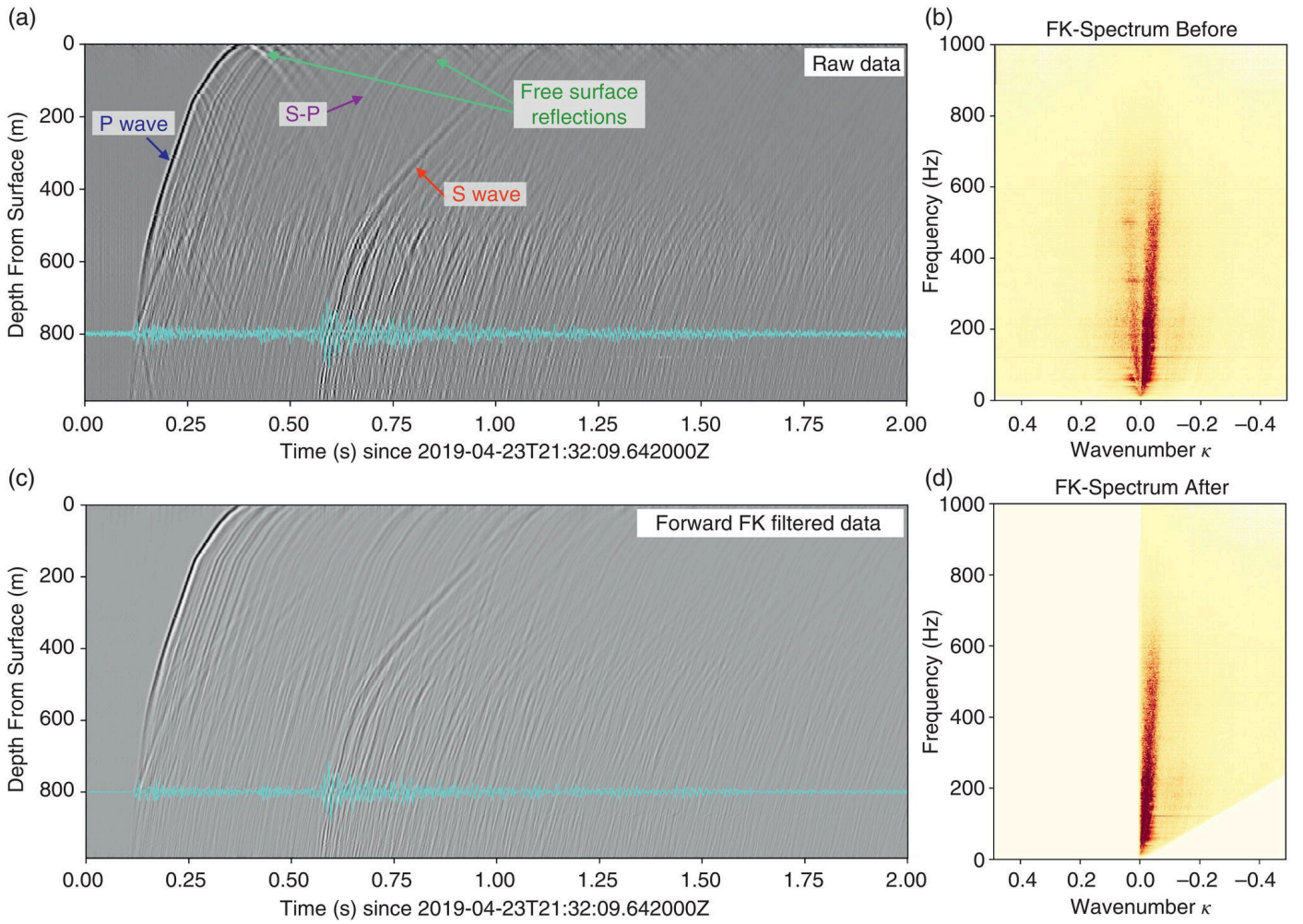


Figure 16.3 Event recorded by the DAS cable installed in Well 78A-32 on 23 April 2019 21:32:09; $M_L = 0.91$. The earthquake was also recorded by the surface array and cataloged by the UUSS. Many different phases can be observed, some of which are labeled with arrows in both the raw data (a) and after a forward FK filter is applied (c): direct P waves (blue), direct S waves (red), their conversion to a P wave at the granite contact (purple), and free-surface reflections (green). In (a) and (c), we also show the waveform for channel 800 of the fiber cable (cyan line). (b) and (d) FK spectrum before and after applying a forward FK filter, respectively.

Figures 16.3 and 16.4 present illustrative examples of 2 s DAS records corresponding to two earthquakes with magnitudes $M_L = 0.91$ and $M_L = -0.8$, respectively. These earthquakes are part of the detection catalog of Lellouch et al. (2021) and belong to the proximal cluster (3–5 km) located ~3 km to the southeast of the stimulation region, hereafter referred to as the “southeast cluster” (Lellouch et al., 2021). Here, we report the magnitude as calculated by Lellouch et al. (2021) using the methodology described in Lellouch et al. (2020). For both earthquakes, it is clear that DAS offers a significant advantage through its quasi-continuous spatial and temporal sampling of the seismic wavefield, where we can recognize distinct arrival patterns as a function of depth along the fiber. Figures 16.3(c) and 16.4(c) show the data for both events after denoising, with their corresponding FK spectra. For the weaker event (Figure 16.4(a)), the energy signal of the raw data is not clearly visible either in the time domain or in the FK spectrum. However, after denoising, the coherent $k = 0$ noise is attenuated as well as the high-frequency random noise, allowing the spatial coherency of the filtered wavefield to be visible in both time and frequency domains (Figure 16.4(c)).

Spectrograms obtained for the two events also exhibit distinct features between the higher and lower magnitude events, which can be directly attributed to the noise levels in each record. The higher-magnitude event displays clear energy peaks corresponding to the P- and S-phase arrivals (Figure 16.5(a)). On the other hand, in the case of the low-magnitude event, although the SNR varies between the individual channels, for most channels we cannot distinguish P or S phases unless a filter is applied (Figure 16.5(b)).

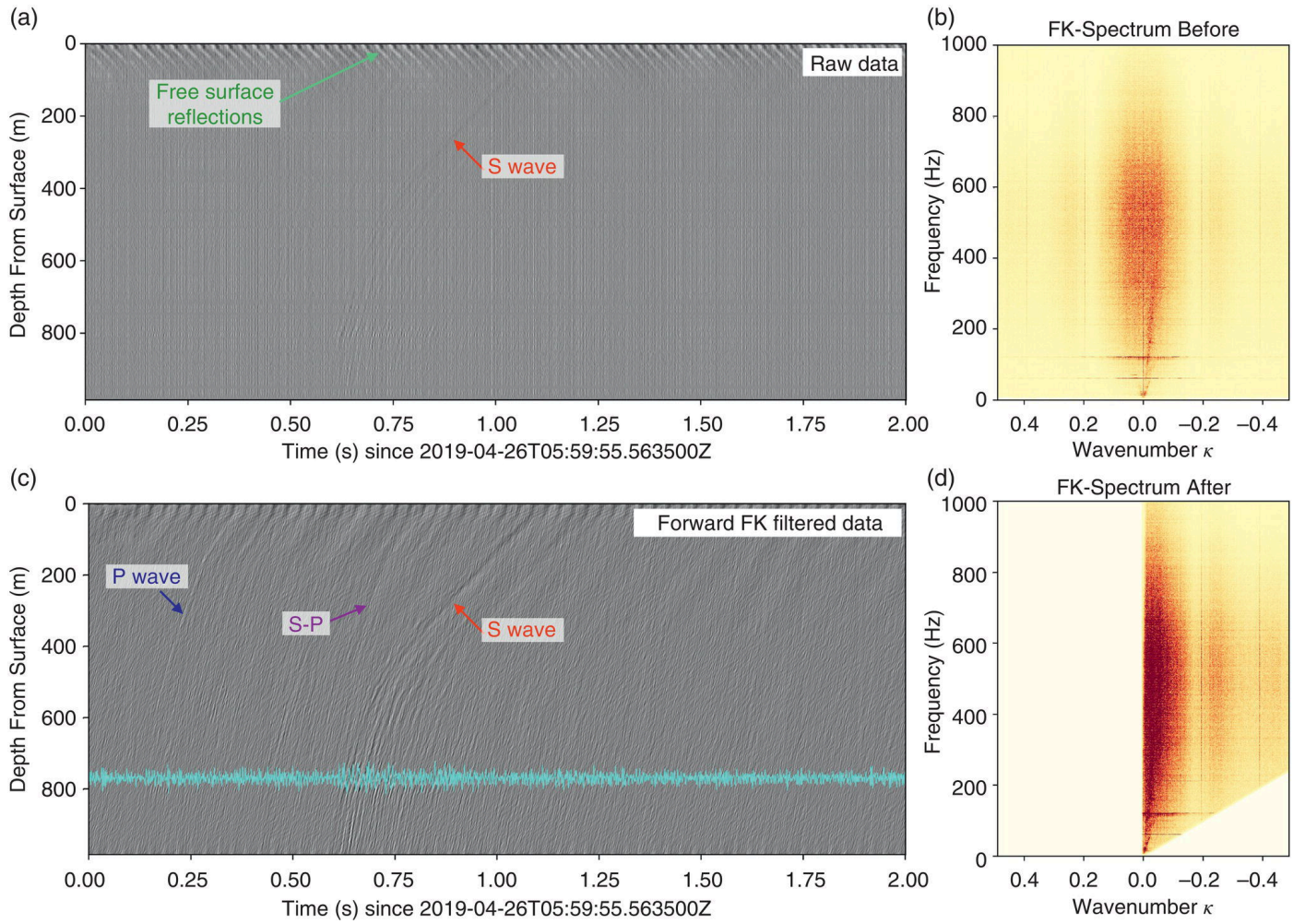


Figure 16.4 Event recorded by the DAS cable installed in Well 78A-32 on 26 April 2019 05:59:55; $M_L = -0.8$. Note how the event phases are hardly visible in the raw data (a), whereas they can be better identified after a forward FK filter is applied (c): direct P waves (blue), direct S waves (red), their conversion to a P wave at the granite contact (purple), and free-surface reflections (green). In (c), we also show the waveform for channel 770 of the fiber cable (cyan line). (b) and (d) FK spectrum before and after applying a forward FK filter, respectively.

DAS-Based Locations

Figure 16.6 shows the DAS-based locations and their uncertainties obtained by applying HADES-R to the southeast seismicity cluster located ~ 3 km from the stimulation patch. Of the 82 earthquakes cataloged by Lellouch et al. (2021), 39 had clearly visible P- and S-phase arrivals that could be picked on the lower 626 m section of the DAS cable, up to a depth of 877 m below the surface. To serve as the master event, we selected a well-recorded earthquake that had been previously located using the surface network, as documented by Lellouch et al. (2021). Similarly to Lellouch et al. (2021), we fix the depth of the master event at 1 km, as the initial location derived with solely surface stations had large depth uncertainties of ~ 3 km. We then repeat the HADES-R location procedure for 100 random channel pairs along the DAS cable and retrieve the distribution of the location uncertainty. We show this in Figure 16.6, where the frequency of earthquake locations found at a specific spatial coordinate by HADES-R over 100 iterations is indicated by the two-dimensional histogram. We observe a certain degree of rotational uncertainty around the master event, with multiple possible solutions for which the rectilinearity is maximized. However, there is a clear tendency for the events to cluster to the southeast of the master event (yellow solid circle in Figure 16.6). The location density distribution agrees with the DAS-derived locations of Lellouch et al. (2021), with the distinction that we are able to obtain a better constraint on depths and a smaller uncertainty density cloud. Both location sets from Tuinstra et al. (2024) and Lellouch et al. (2021) show a strong correlation with the historical seismicity underlying the Mineral Mountains.

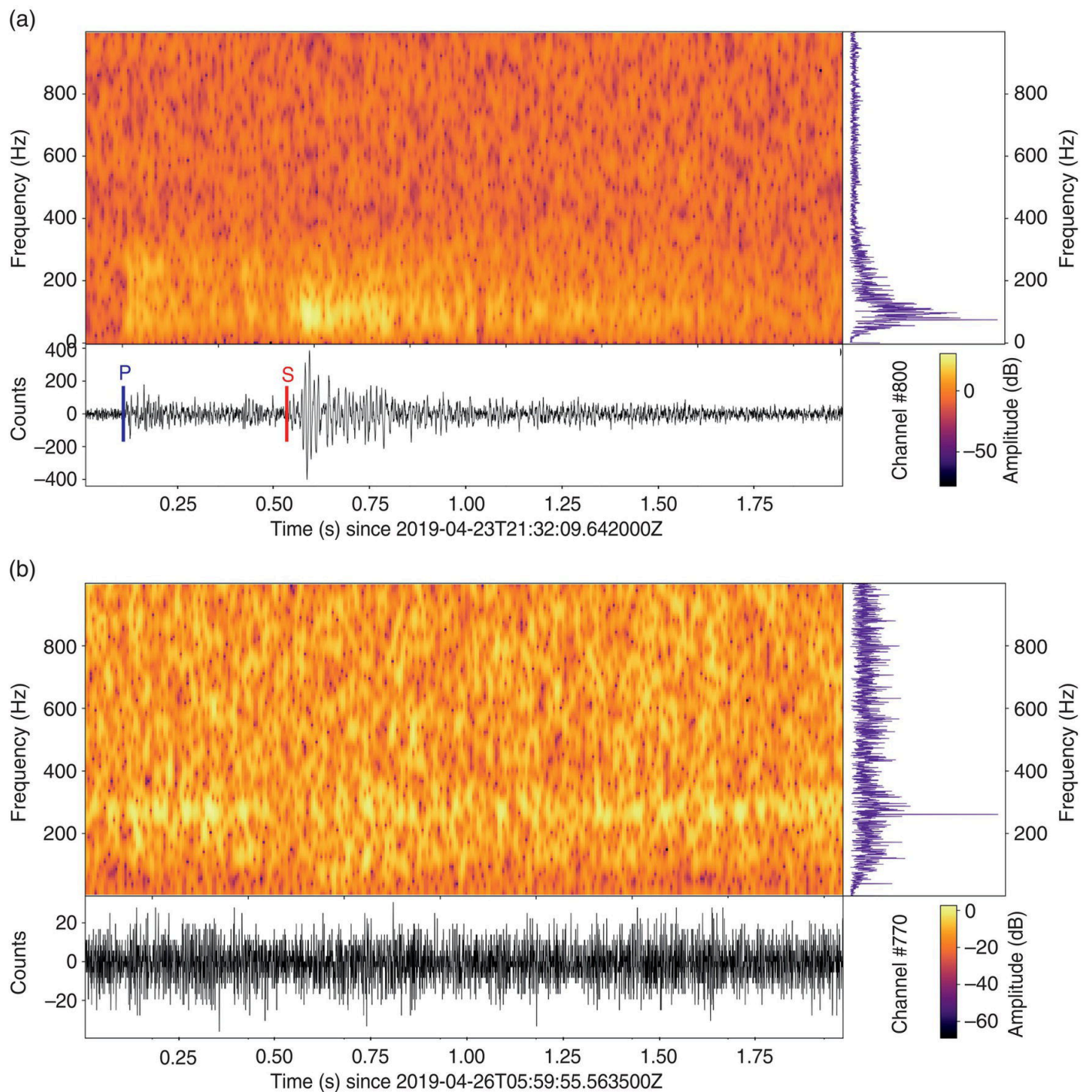


Figure 16.5 Spectrograms for a single channel of the fiber-optic cable installed in Well 78A-32. (a) Event 2019-04-23 21:32:09; $M_L = 0.91$; DAS channel 800. The bottom panel shows the raw seismogram recorded by DAS channel 800, and the right panel indicates the corresponding amplitude spectrum. (b) Event 2019-04-26 05:59:55; $M_L = -0.8$. DAS channel 770. The bottom panel shows the raw seismogram recorded by DAS channel 770, and the right panel indicates the corresponding amplitude spectrum.

Adapted from Lellouch et al. (2021).

16.3.2 Bedretto Underground Rock Laboratory

Whereas FORGE offers a testbed for experiments at the full-reservoir scale, the Bedretto Underground Laboratory for Geosciences and Geoenergies (“BedrettoLab” hereafter) provides the ideal environment for controlled mesoscale in situ experiments to further improve our understanding of hydro-seismo-mechanical processes taking place in fractured crystalline rock masses (Ma et al., 2022).

The BedrettoLab is located in the Bedretto Tunnel in the Swiss Central Alps, near the Gotthard Pass (Figure 16.7a). With a length of 5,218 m, the tunnel connects the Furka Base Tunnel in the northwest to the Bedretto Valley in the southeast and was originally excavated to transport muck as part of the Furka Base Tunnel construction logistics (Keller & Schneider, 1982). Since 2018, the Bedretto Tunnel has been made available by its owner, Matterhorn Gotthard Bahn (MGB), to ETH Zurich to perform experimental research focusing on the behavior of the deep underground in various domains, including geothermal energy and earthquake physics.

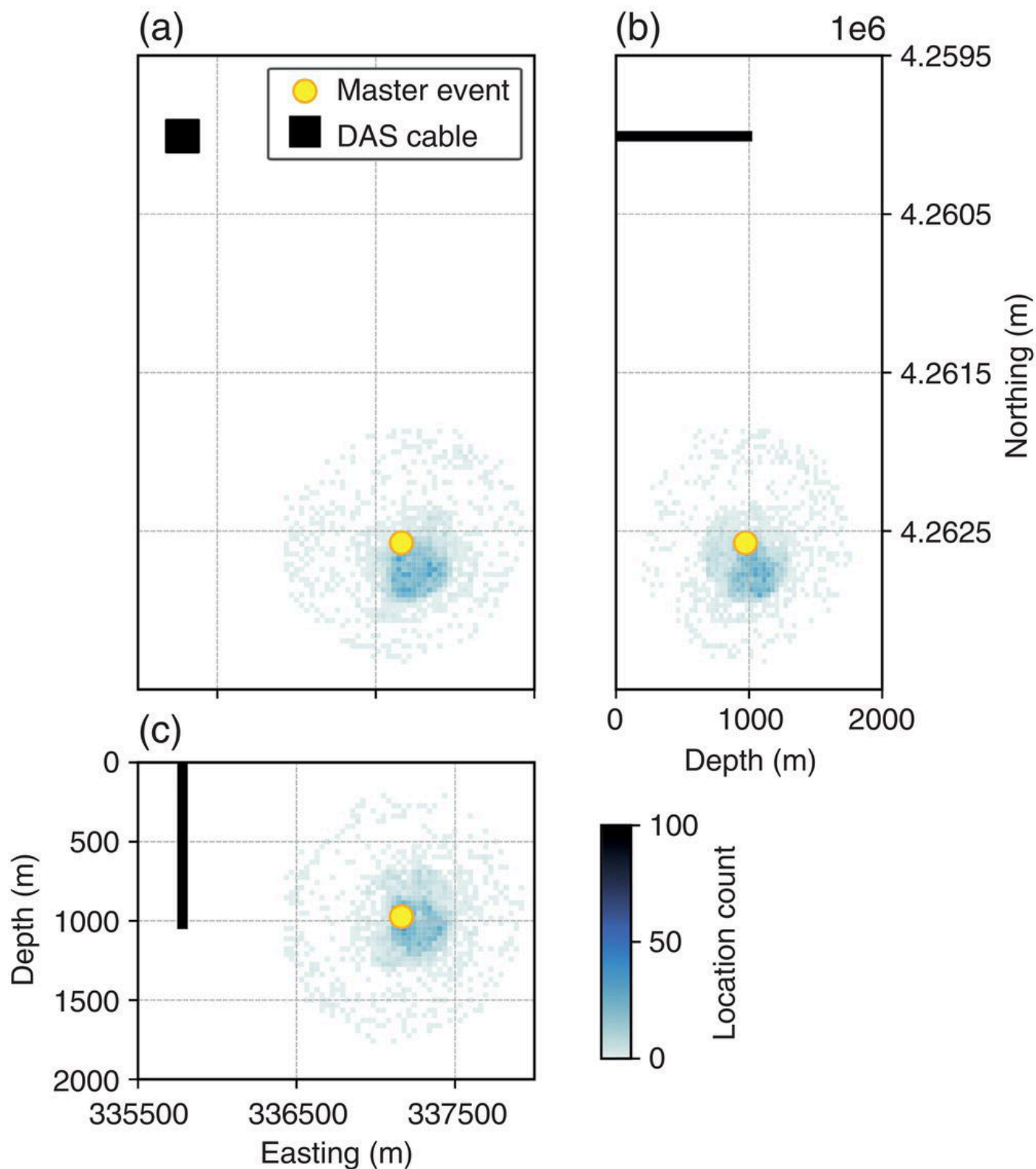


Figure 16.6 HADES-R derived locations of the southeast seismicity cluster recorded by a vertical downhole DAS during the 2019 FORGE pilot stimulation in the horizontal (a), longitude-depth (b), and latitude-depth plane (c). The master event is indicated by the yellow solid circle, and the uncertainty cloud is shown as a two-dimensional histogram (bluish colors).

Adapted from Tuinstra et al. (2024).

Since then, several boreholes along the tunnel have been excavated and instrumented for detailed geophysical characterization of the experimental site. The goal is to provide a better understanding of the potential for geothermal energy by researching hydrofracturing via various injection strategies in a mesoscale, controlled environment close to the injection point (Plenkers et al., 2023). Several injection activities have been conducted at the BedrettoLab, with the first operations in the Fall of 2020 (Hertrich et al., 2021) and more occurring in 2022/2023 (Broeker et al., 2023).

Here, we focus on a seismicity cluster that was observed in May 2021 about 900 m from the instrumented laboratory, toward the Bedretto Valley (Figure 16.7). At the time of occurrence of the cluster, the seismic network inside the tunnel was composed of the

following: three seismic stations along the tunnel itself; an eight-level three-component (3C) digital geophone chain in borehole ST1; an Avalon single 3C analog geophone with downhole amplifier in borehole ST1; and a single 3C IMS analog geophone in one of the monitoring boreholes. Borehole ST1, which is slightly deviated, features a multipacker system (14 intervals in total) for a total length of about 400 m, with a hybrid cable installed through the packer system. This hybrid cable, manufactured by Solifos and referred to as BRUsens Heatable, features distributed fiber-optic sensing with an actively heated element. The fibers are installed in a metal tube (FIMT) with two SM and two MM loose fibers. Active heating can be achieved using copper conductors. The MM fibers are looped at the end of the boreholes and were used for monitoring of distributed temperature (DTS), and the SM fibers are both single ended, both used for injection monitoring with DAS and long-term monitoring with distributed strain sensing (DSS) interrogators.

The dataset employed here was recorded with an iDASv2 that recorded continuously in the period 24 April to 10 May. DAS data were acquired with a 1 m channel spacing using a 10 m gauge length and a sampling rate of 1 kHz, with 20-fold internal stacking. The seismicity cluster contains 17 events detected in the period between 24 April and 10 May that were identified in the DAS dataset. [Figure 16.7](#) shows the location of a subset of 11 events. These events were additionally detected and located using the geophone array, which was, however, continuously recording only from 2 to 9 May by Geo-Energie Suisse (GES). For its location estimates, GES used P- and S-wave arrival times, as well as hodogram azimuths from the 3C eight-level geophone chain and L2 minimization approach over a search grid of 1 m. Despite the fact that V_p and V_s velocities were not optimized, the direction of the events was constrained by hodograms analysis and resulted in an uncertainty of $\pm 2^\circ$. The relative location resolution was estimated to be between 80 and 90 m in depth and between 20 and 30 m in the horizontal plane. Of the recorded events, the largest magnitude $M_L = 0$ was clearly detected by the routine processing of the regional surface network and reported in the Swiss Seismological Service (SED) catalog ([Figure 16.8](#)).

The cluster of events observed in the DAS occurred before and after some experiments conducted by GES featuring injections in borehole ST1, which generated little to no seismicity in the instrumented reservoir. Similarly to our first case study at the FORGE site, we refrain from interpreting the origin of the cluster recorded during the 24 April to 10 May period and focus here on the location of events and analysis of the seismic recordings of the fiber optics cable installed in borehole ST1.

[Figure 16.8](#) illustrates an example of a 0.6 s DAS record for the largest event with magnitude $M_L = 0$, as reported by the SED catalog. We caution here that the magnitude calculated by the national network may be overestimated. In its estimation, GES followed a time-domain approach over a window of 20 ms from the P- and S-wave first arrivals picked on the geophone chains and thus may be better calibrated with respect to that of the national network. For the same event shown in [Figure 16.8](#), GES calculated a $M_w = 0.27$. As we have seen at the FORGE site, we can apply FK filters to increase the SNRs of small, weak events ([Figure 16.8\(c\)–\(e\)](#)). In this case, we also observe a strong scattering effect of the first arriving phases, which could be because the fiber passes through the packer systems installed in the same borehole ST1. [Figure 16.8\(c\)–\(e\)](#) show various filters that were tested to attenuate these scattering effects. With a bidirectional filter that allows only high apparent velocities, the scattering noise is much attenuated but cannot be removed completely. When we apply a one-directional filter, it seems to remove some broadside energy that could be linked to reflections within the packer systems, although we may also lose some signal information. For P- and S-phase picking tasks, we opted for the more standard bidirectional FK filter as we can still observe quite strong P- and S-wave phases ([Figure 16.8\(c\)](#)), also clearly visible in the spectrogram ([Figure 16.9](#)).

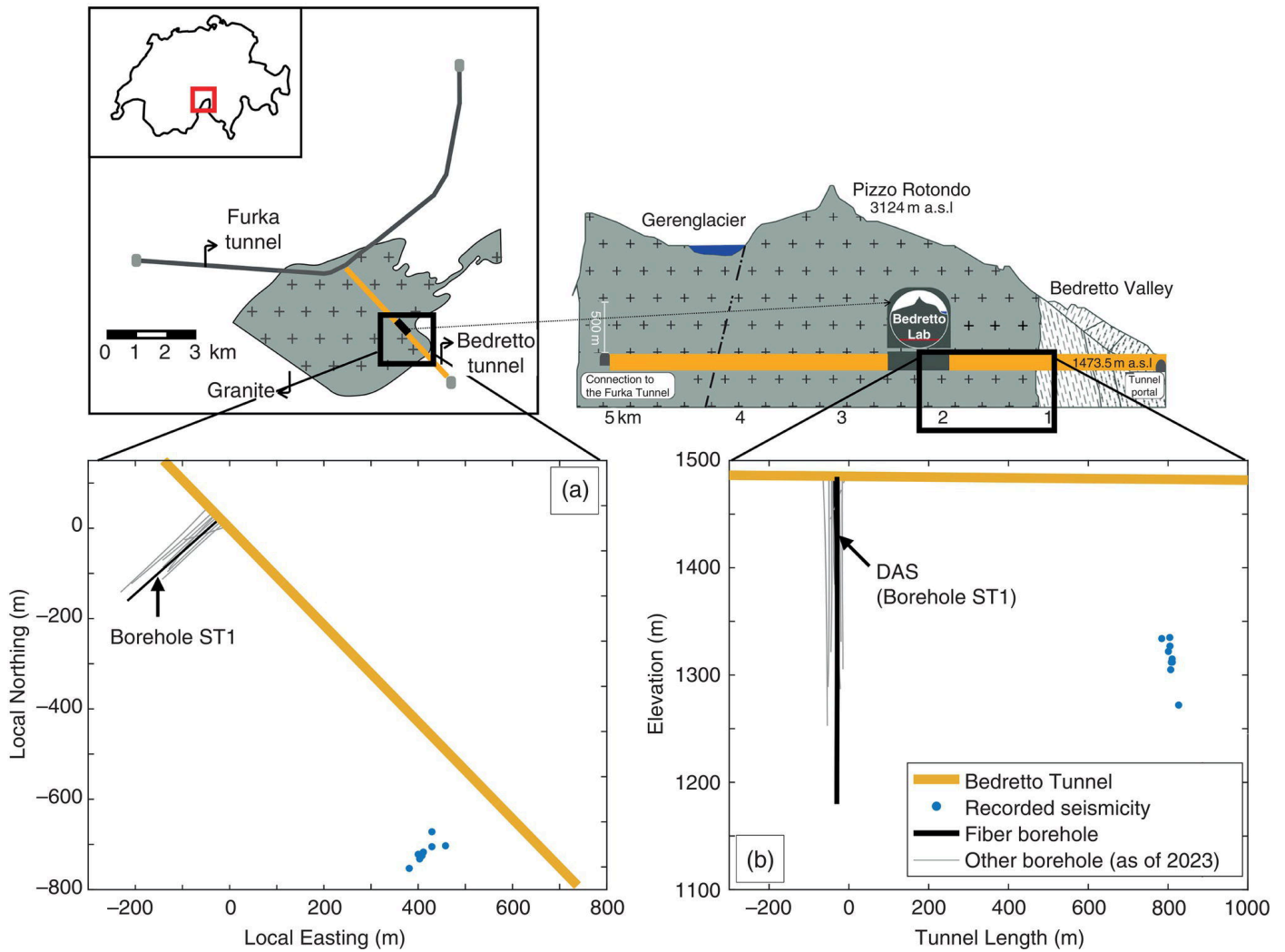


Figure 16.7 Overview map of the BedrettoLab, in Switzerland, in plane view (a) and cross-section (b). The Bedretto Tunnel is shown by the orange line, and all the boreholes drilled as of 2023 are shown with gray lines. The thick black line indicates borehole ST1 instrumented with DAS.

DAS-Based Locations

Similar to the southeast cluster at FORGE, we applied HADES-R to the distant seismicity cluster recorded between 24 April and 10 May 2021. On the DAS record, of the 17 events detected, 14 were suitable to pick both the P and S phases and thus were used in the HADES-R workflow. The master event was selected among the events located using the geophone chains by GES.

Figure 16.10 shows the DAS-based locations obtained by applying HADES-R to the cluster. In this case, we repeat the location procedure for 150 random channel pairs along the DAS cable and retrieve the distribution of the uncertainties. The location density cloud agrees quite well with the geophone-derived locations of the GES catalog. The depth uncertainties observed are also consistent with those reported by GES, which similarly ranged in the hundreds of meters. Here, the presence of a nonvertical monitoring well, and thus a better angular configuration between stations and events, significantly reduces the ambiguity in the azimuthal direction, leading to a reduction in nonuniqueness associated with the rotations performed to find the absolute location of the cluster from its relative initial position. As evident from the two-dimensional histograms, a distinct preferred orientation of the cluster in the NE–SW direction, rather than a circular pattern, becomes apparent.

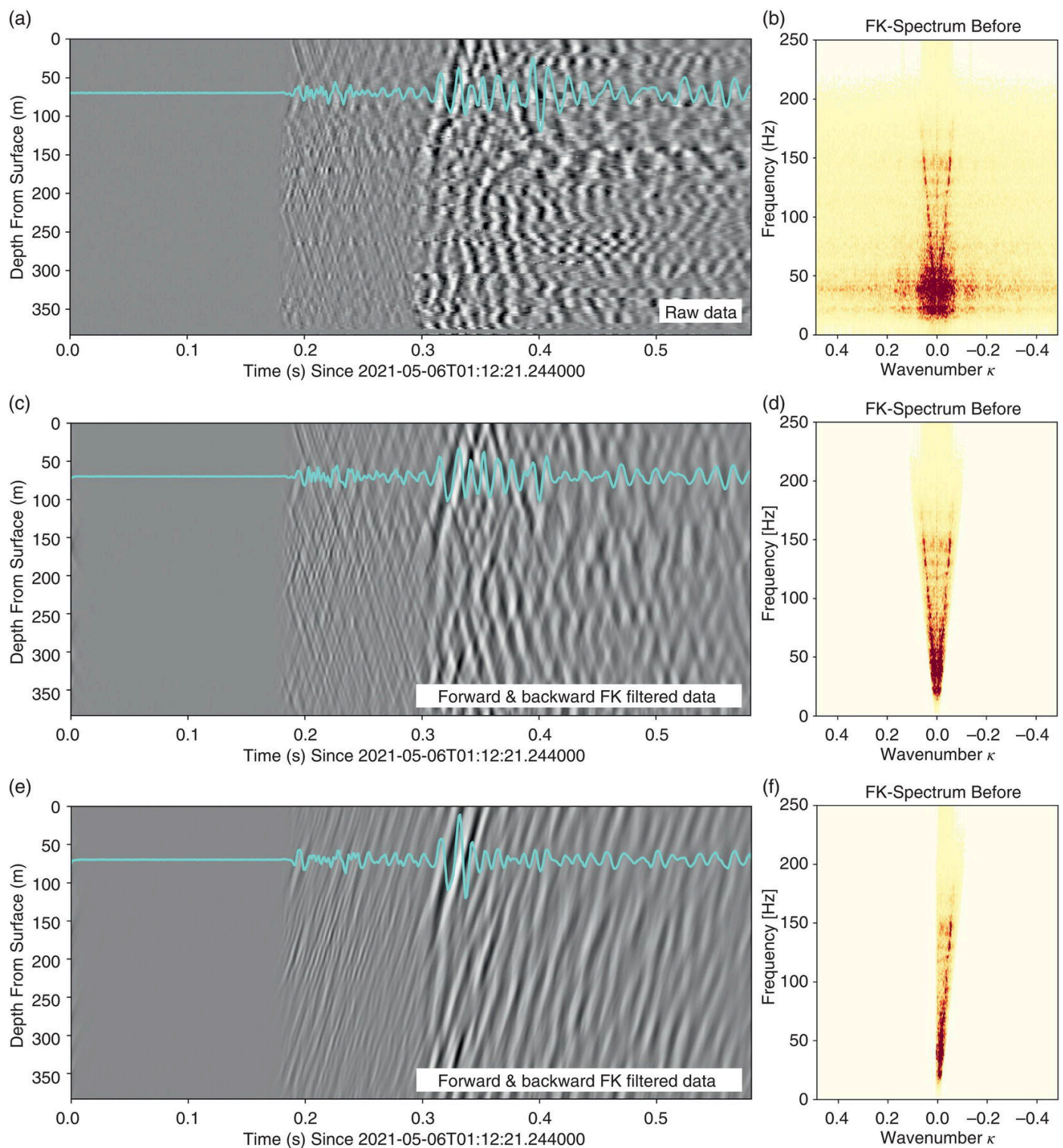


Figure 16.8 Event recorded by the DAS cable installed in borehole ST1 on 6 May 2021 01:12:21; $M_L = 0.0$ from the SED catalog solution ($M_w = 0.27$ as calculated by GES). Note the strong scattering effects due to the fiber passing through the packer system. (a) Raw data; (c) filtered data following a forward and backward bidirectional FK filter; (e) filtered data following a forward and backward one-directional FK filter; in (a), (c), and (e), we also show the waveforms for channel 70 of the fiber cable (cyan line). (b), (d), (f) FK spectrum before and after applying forward and backward FK filters, respectively.

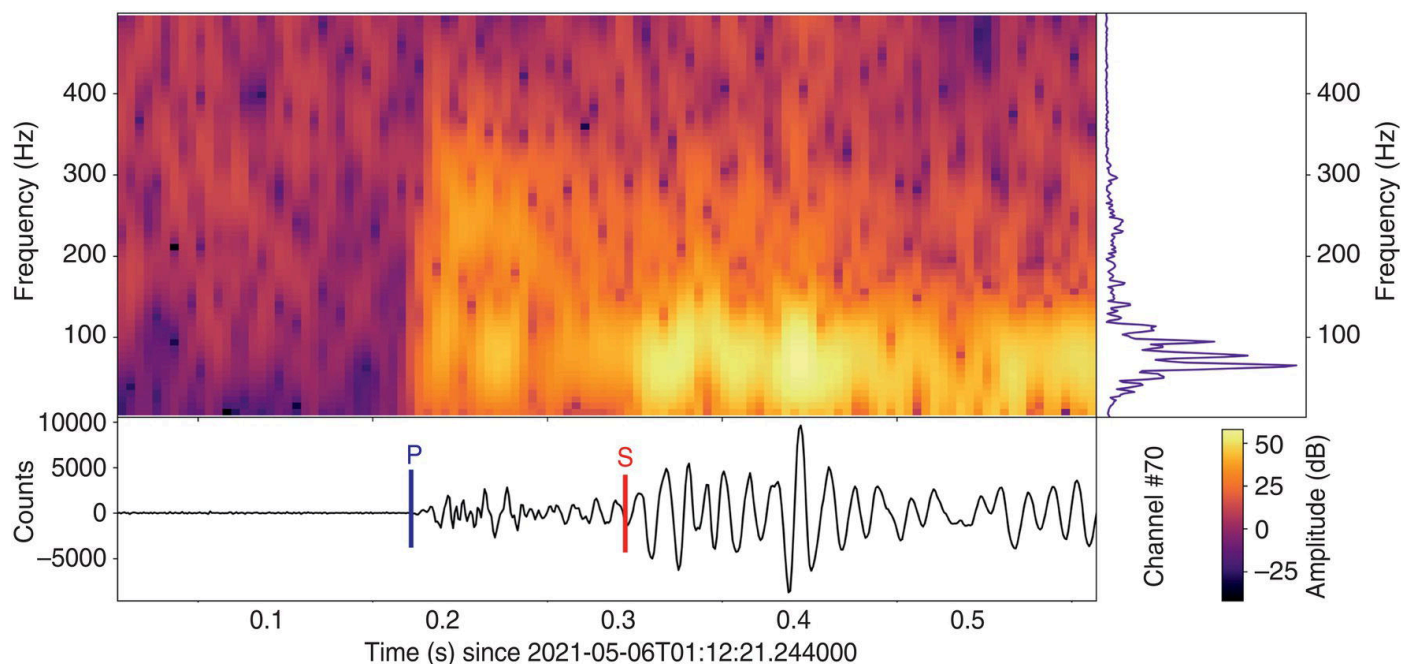


Figure 16.9 Spectrograms for channel 70 along the fiber. The bottom panel shows the raw seismogram recorded by DAS channel 70, and the right panel indicates the corresponding amplitude spectrum.

16.4 DISCUSSION

The two showcased examples at FORGE and BedrettoLab demonstrate the capability of downhole DAS technologies to effectively detect and locate low-magnitude earthquakes and microseismic events, even at distances of several meters to kilometers from the respective monitoring wells where the DAS systems are deployed. It also shows that DAS technologies are well suited for passive monitoring in both the far and near fields across different scales, ranging from low-magnitude events ($-4.0 < M < 0.0$) detected in an underground field laboratory to hydraulic fracturing-induced microseismicity ($-2.0 < M < 0.5$), to local earthquakes ($M > 1.0$).

Detectability capabilities of DAS systems are somewhat hindered by the inherently noisier nature of single channels of DAS data with respect to conventional downhole geophone arrays (Correa et al., 2017). However, we have shown that traditional preprocessing filtering routines and FK filters can already perform quite well. We see this clearly at the FORGE site, where the wavefield phases become visible in both the frequency and time domains after the application of FK filters (Figure 16.4(c)). Yet certain types of noise cannot always be eliminated through conventional filtering techniques. This is observed in the BedrettoLab dataset, where we are unable to separate the signal from the scattering. We argue that such an effect could be produced by the presence of packer systems in the well and the fiber-optic cable feeding through the packers (more details about the well can be found in Plenkers et al., 2023). For these cases, more advanced techniques may be needed, such as deep learning methods (e.g., Jousset et al., 2022; Schwarz, 2019; van den Ende et al., 2021) or adaptive filtering (e.g., Isken et al., 2022). Although the performance of downhole geophones is undoubtedly superior to single-channel downhole DAS in terms of recovered SNR, if we compare the DAS records with surface stations, it becomes apparent that the deeper installation of downhole DAS renders it unaffected by both anthropogenic noise and wind noise, as well as by the dissipative heterogeneous shallow layers of the Earth (Hutt et al., 2017; Lellouch et al., 2021). DAS deeper deployments at reservoir depths are also possible due to the DAS robustness to high temperatures and pressure. As DAS cables can be installed permanently and in a cost-effective way in boreholes (both monitoring and stimulation ones), they offer a promising alternative for long-term passive monitoring across a wider area beyond the immediate vicinity of the stimulation patch. At FORGE, Lellouch et al. (2021) were able to improve the magnitude completeness of 0.5–1.0, well into negative magnitudes (as also observed in BedrettoLab), even though the well was situated 3–4 km from the nearest seismically active region.

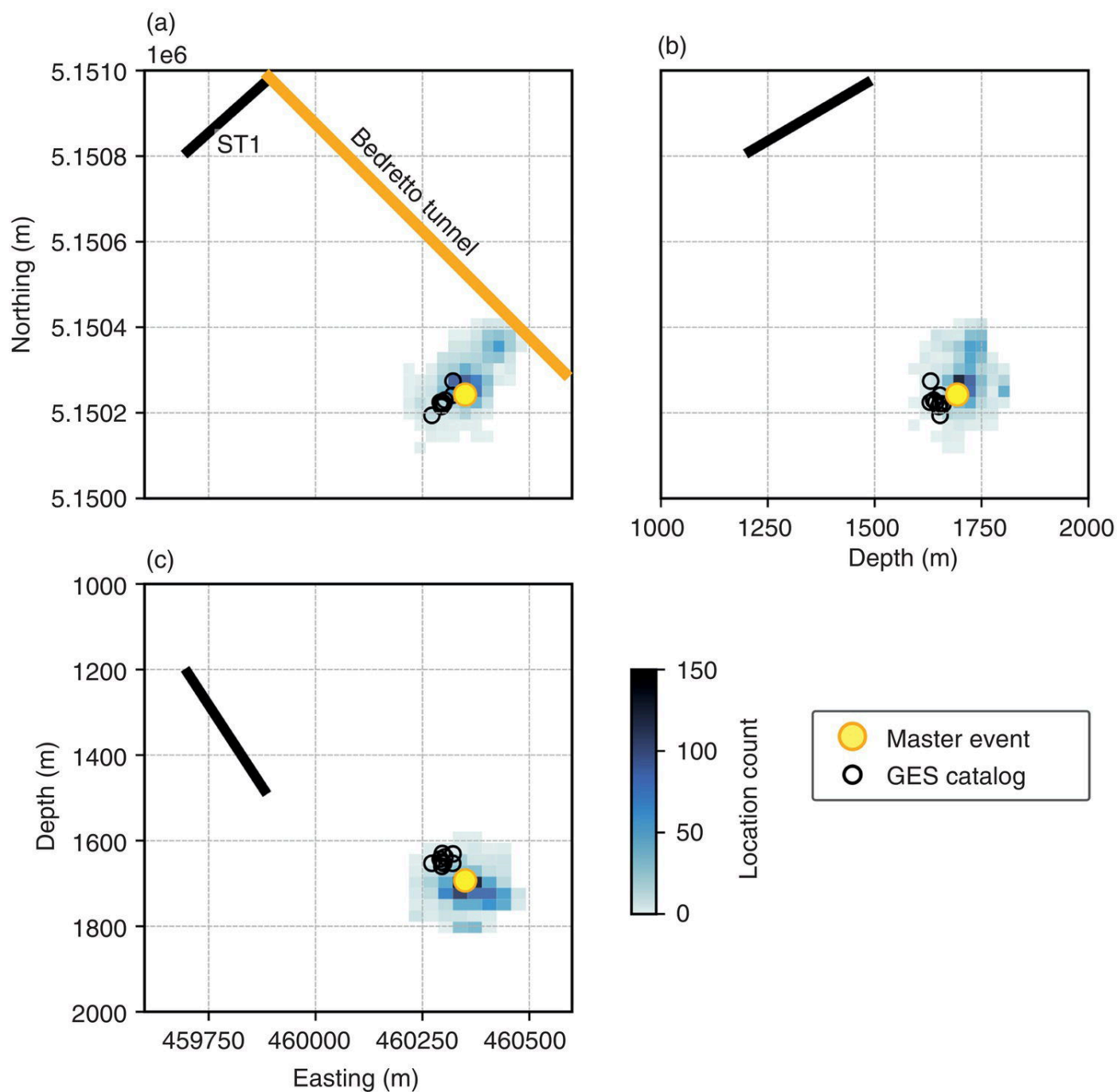


Figure 16.10 HADES-R derived locations of the southeast seismicity cluster recorded by the downhole DAS array in borehole ST1 in the horizontal (a), longitude-depth (b), and latitude-depth plane (c). The master event is indicated by the yellow solid circle, and the uncertainty cloud is shown as a two-dimensional histogram (bluish colors). Open black circles indicate the locations derived by GES using colocated geophone chains.

In settings where geophones might experience mechanical issues due to elevated temperature conditions or where surface noise obstructs the identification of deep, low-magnitude events, event localization using DAS-only network configurations has gained traction in microseismic monitoring applications. In terms of DAS technology's locatability potential, the inherent azimuthal ambiguity of a single vertical downhole DAS cable poses a limitation on the directional resolvability of DAS-based locations. However, we have shown that the HADES-R algorithm can retrieve the general shape of a seismic cluster if the distance between the receivers and the cluster centroid is much larger than the overall size of the cluster and at least one event in the cluster has a well-known absolute location. These assumptions are often valid for the illustrated cases of out-of-network seismicity.

Additionally, HADES-R can deliver information on location uncertainties by leveraging the DAS channel abundance through an iterative approach in which the cluster orientation uncertainty cloud is computed. In extremely ill-posed network configurations (e.g., single vertical well) and where clustered seismicity occurs out-of-network, Tuinstra et al. (2024) demonstrated that by incorporating a subset of events that are also visible on surface stations, a better estimate of the cluster shape and network aperture can be achieved. Similarly, incorporating 3C geophones can be effective in reducing the azimuthal ambiguity as information derived from hodogram analysis can provide crucial a priori azimuthal constraints. This highlights how additional instruments can improve location estimates. This approach of merging dynamic strain sensing and seismic data has also been tested and demonstrated by Flóvenz et al. (2022) and Binder and Abatchev (2021). With the Bedretto case study, we also show that even a slightly deviated well

could improve the azimuthal ambiguity and provide a reliable DAS-based only location with minimal prior information and no assistance from a surface network.

DAS-based locations provide a more accurate depth estimation compared to the surface array. Thus, network geometry is also important and strongly affects the location capability of DAS. A generalization requires specific forward simulations, and very often, given the cost to drill boreholes, their geometry is driven by other factors (e.g., to maximize energy exploitation). From a seismological point of view, the drilling of a horizontal or inclined well enhances the azimuthal gap, and it is always favored over a vertical borehole.

Although this is also true for any other type of borehole array, fiber-optic cables provide several advantages with respect to conventional sensor arrays, as detailed previously. Hence, we believe that downhole DAS arrays, even when deployed in a single vertical well, can serve as a valuable complement to existing local networks, whether in boreholes or at the surface.

One further outcome from our evaluation is the ability of DAS to perform equally across scales. DAS signal characteristics and location capabilities are tested at the FORGE site (full-scale reservoir) and BedrettoLab (mesoscale reservoir), with results showing overall good performance. This has favorable impacts on the usage of downhole DAS in vertical or in deviated wells, at close or far distances, for high- or low-magnitude events. Nonetheless, we must emphasize that acquiring more extensive data through longer experiments is crucial to obtain a more robust statistical understanding of DAS capability to complement conventional earthquake monitoring network arrays.

16.5 CONCLUDING REMARKS

DAS offers numerous operational advantages for microseismic monitoring: it can be permanently installed in extreme environments (e.g., high pressure and temperatures), can remain in place undisturbed behind casing, and can interrogate long distances with high-density spatial and temporal coverage. Here, we have shown two case studies at the mesoscale and full-scale reservoir to evaluate the potential of downhole DAS to detect and locate out-of-network microseismicity. Our results at FORGE and BedrettoLab indicate that with adequate preprocessing, downhole DAS data quality allows the detection of very low-magnitude earthquakes several meters or kilometers from the well. HADES-R is able to retrieve the shape of out-of-network clustered seismicity across scales, produce more accurate depths, and provide uncertainty distribution. To avoid azimuthal ambiguity, even a slight deviation of the wells can greatly aid in the recovery of the correct rotation for obtaining robust locations. Our evaluation supports the employment of longer-term experiments to keep exploring downhole DAS suitability for continuous passive microseismic monitoring.

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AVAILABILITY STATEMENT

Both geophone data and DAS data at FORGE are openly available for download at the Geothermal Data Repository (GDR) website (<https://gdr.openet.org/forge>). DAS data acquired in May 2021 at the BedrettoLab from borehole ST1 are available upon request.

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